COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 5 : SLO 2 — Booth's Algorithm (Simulation)
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1. Terminology Check (Remembering)

• 1. M → b. The register that holds the Multiplicand
• 2. Q → c. The register that holds the Multiplier
• 3. Q-1 → a. The bit to the right of the LSB of Q, initialized to 0
• 4. A → d. The register that holds the partial product, initialized to 0

2. Explaining the Steps (Understanding)

• If the bits are 10: Subtract M from A (A = A - M), then perform an arithmetic right shift.
• If the bits are 01: Add M to A (A = A + M), then perform an arithmetic right shift.
• If the bits are 00 or 11: perform only the arithmetic right shift (no arithmetic operation).

3. Positive Multiplication Simulation (Applying): 5 × 3, M = 0101, Q = 0011

• Initial: A = 0000, Q = 0011, Q-1 = 0
• Cycle 1: Q0Q-1 = 10 → A = A - M = 1011; shift → A = 1101, Q = 1001, Q-1 = 1
• Cycle 2: Q0Q-1 = 11 → no operation; shift → A = 1110, Q = 1100, Q-1 = 1
• Cycle 3: Q0Q-1 = 01 → A = A + M = 0011; shift → A = 0001, Q = 1110, Q-1 = 0
• Cycle 4: Q0Q-1 = 00 → no operation; shift → A = 0000, Q = 1111, Q-1 = 0
• Final Answer (A, Q): 0000 1111 = +15 (5 × 3 = 15 ✓)

4. Negative Multiplication Simulation (Applying): 4 × -3, M = 0100, -M = 1100, Q = 1101

• Initial: A = 0000, Q = 1101, Q-1 = 0
• Cycle 1: Q0Q-1 = 10 → A = A - M = 1100; shift → A = 1110, Q = 0110, Q-1 = 1
• Cycle 2: Q0Q-1 = 01 → A = A + M = 0010; shift → A = 0001, Q = 0011, Q-1 = 0
• Cycle 3: Q0Q-1 = 10 → A = A - M = 1101; shift → A = 1110, Q = 1001, Q-1 = 1
• Cycle 4: Q0Q-1 = 11 → no operation; shift → A = 1111, Q = 0100, Q-1 = 1
• Final Answer (A, Q): 1111 0100 = -12 in 2's complement (4 × -3 = -12 ✓)

5. Find the Mistake (Analyzing): 2 × -2, M = 0010, Q = 1110

• Mistake: the bits Q0 Q-1 are 10, so the correct action is A = A - M, not A = A + M as written in Cycle 1.
• Corrected Cycle 1 row: M = 0010, A = 1110 → after shift A = 1111, Q = 0111, Q-1 = 0; Action: A = A - M (A = 1110), then arithmetic right shift.